

BS-SiamRPN: Hyperspectral Video Tracking based on Band Selection and the Siamese Region Proposal Network*

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Abstract—The spectral information of hyperspectral data is helpful to improve the performance of target tracker. Limited by the number of training samples, most video tracking methods based on deep learning cannot effectively utilize the semantic information of hyperspectral data. Therefore, this paper proposes a hyperspectral video target tracking method based on Band Selection and the Siamese Region Proposal Network (BS-SiamRPN). Firstly, an intelligent optimization algorithm is utilized to determine three bands with the largest joint entropy. Based on limited hyperspectral information, Transfer Learning (TL) is performed on the pre-trained siamese networks to obtain a good parameter. Here, index. Then, the dataset in Generic Object Tracking (GOT) benchmark is utilized to retrain the SiamRPN method. Finally, the information of three specific bands referring to hyperspectral videos is regarded as the input of the migration network, which can obtain matching results. Experimental results on public hyperspectral data show that the proposed siamese network based tracking method effectively utilize spectral features, and do well in visual effects and objective evaluation.

Index Terms—Target tracking, Hyperspectral video, Siamese network, Transfer learning

I. INTRODUCTION

Target tracking on Hyperspectral Videos (HSV) is to give the state information of the target at the initial moment of hyperspectral videos, and continuously evaluate the target's state at the subsequent moment. For classical visual target tracking methods, it is fatal to distinguish the target and the background with similar color. Further, the hyperspectral video tracking methods have the advantage of taking into account the spatial and spectral information of the object, which helps to deal with the excursion during tracking effectively [1]. However, the large amount of data affect the real-time performance of the Deep Learning (DL) based trackers. Besides, the redundancy of spectral data needs to be eliminated, in order to improve the robustness of the algorithm.

Recently, tracking methods mainly includes DL based generative methods [2]–[8] and Discriminative Correlation Filtering (DCF) based methods [9]–[16]. The DCF method [17], [18], whose goal is to obtain a correlation filter that denotes the

inherent difference between the target and its neighborhood, trains filters efficiently in the frequency domain. Bolme et al. proposed the fast Minimum Output Sum of Squared Error (MOSSE) tracker [9], which uses gray-scale features and achieves a high speed. Further, some improvements adopt other features, such as Histogram of Gradient (HOG) features [10], color name features [19], and DL features [5]. The Kernelized Correlation Filter (KCF) method [10] exploits the advantage of multi-channel HOG features with the kernel trick. Wang et al. presented a multi-Cue correlation based tracker [20] with multifeature of HOG and VGG (Visual Geometry Group). These improved DCF trackers attempt to solve the occlusion problem [21] of the MOSSE method. Nevertheless, these discriminative methods are also sensitive to the false target with color similarity.

The DL tracking focuses on the extraction of deep features, such as the Deep Learning Tracker (DLT) [2] and siamese network tracking [4], [6], [22], [23] methods. Generally, it introduces the offline training with online tracking, which extracts objects' inherent features and updates the parameter. Several works, such as the fully Convolutional Neural Networks (CNN) tracking [3] method and the Siamese Region Proposal Network (SiamRPN) [24] tracking method, have combined DL with the DCF based tracking framework. Instead of obtaining hand-crafted features, the DCF trackers adopted features automatically learnt via CNN, which improves the robustness of the tracking significantly. Zhang et al. proposed the lightweight CNN based tracker [25] without training process. However, the accuracy of the above visual tracking methods is easily affected by similar color in the background. Besides, independent Graphics Processing Unit (GPU) is needed to computing deep features in most cases.

Based on the above analysis, the tracking method of deep learning combined with correlation filtering has been widely applied to single spectrum image processing of visible light and infrared, and achieved better performance [26]–[28]. In visible videos, the color of the target is similar to that of the background, which results in the tracking excursion problem. Therefore, the spectral characteristics of hyperspectral video lay a foundation for solving this problem. At present, scholars

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have provided some effective hyperspectral video tracking methods [29]–[31]. Qian et al [1] applied the convolution network without training to this data set for the first time, and the tracking framework of correlation filtering also improved the execution efficiency of the algorithm. However, this method does not consider eliminating the redundancy of spectral information. Feng et al utilized the hyperspectral unmixing method to obtain features with strong classification performance. This method also adopts the 3D HOG model into the KCF tracking framework, and achieve good tracking results. Besides, Li et al. [29] proposed BaeNet, which used band selection network to input band groups into VITAL [32] tracker for tracking. However, due to the limited number of hyperspectral videos, the training of this method is not sufficient.

Specifically, most hyperspectral video tracking methods are based on extracting traditional features, and do not consider the popular end-to-end network method. Therefore, this paper applies a deep Siamese Network to hyperspectral data processing, and uses the advanced semantic features to estimate the target state. In fact, the whole framework includes band selection model, network training and feature migration. The first part is based on the intelligent optimization algorithm, which obtains the optimal three bands referring to the joint entroph or the Optimum Index Factor (OIF) index. Then, The RGB data from Generic Object Tracking (GOT) benchmark [33] is utilized for network pre-training of the Siamese Region Proposal Network. In order to extract the feature, the hyperspectral data are used for Transfer Learning (TL) [34]. Finally, the three-band information is sent to the feature extraction network, and the semantic information of the object is obtained. Several tracking results are shown in Fig. 1. As a matter of fact, the proposed tracker without TL is sensitive to appearance of objects.

The relevant content of this article is as follows. In section 1, the related work is introduced. In Section 2, the specific content of the proposed tracking algorithm is given. Then, the simulation results and analysis of the siamese network method and comparison algorithm on hyperspectral video are presented in Section 3. Finally, a conclusion is drawn in Section 4.

II. THE PROPOSED TRACKING ALGORITHM

The number of training samples for the public hyperspectral videos is limited, which leads to insufficient semantic information of images acquired through training. Benefit from the twin SiamRPN [24], the semantic information of hyperspectral images is obtained by transfer learning, which improves the performance of the tracking algorithm.

A. SiamRPN

SiamRPN is an end-to-end offline training network based on large-scale image pairs, which transforms the original similarity calculation problem into the problem of regression or classification. In the SiamRPN algorithm, feature extraction is accomplished via a siamese network, which is consistent

with the structure of Fully-Convolutional Siamese Networks (SiamFC) [35]. Besides, the regional proposal network is composed of regression and classification branches. Specifically, the former is used to calculate the Intersection Over Union (IOU) of the output box versus the real target, and distinguish the target from the background with a threshold. The regression branch fine-tunes the output target candidate box to make it closer to the real target state. The existence of regression branches enables the algorithm to ignore the original scale variation.

Although tracking algorithms, such as CFNet [36], DSiam [37] and SiamRPN++ [2], balance the accuracy and the speed, some problems still need to be solved. First, most target tracking algorithms based on siamese networks use features that can only distinguish foreground from non-semantic background. Semantic background is usually considered to be a disturbing factor. When the background is cluttered, the performance of tracker cannot be guaranteed. Second, siamese trackers lack an online update model. The tracker drifts to the background easily, in the case of that the appearance of the target changes dramatically. Third, the local search strategy in the siamese tracker is sensitive to the challenge of the target being partially blocked. However, hyperspectral video contains rich spectral information which can make up for the defect of visible light data to some extent.

The classification branch and regression branch carry out convolution operation on the features of template frame and detection frame respectively,

$$A_{w \times h \times 4k}^{cls} = \phi(x)_{cls} * \phi(z)_{cls} \quad (1)$$

$$A_{w \times h \times 4k}^{reg} = \phi(x)_{reg} * \phi(z)_{reg} \quad (2)$$

where $A_{w \times h \times 4k}^{cls}$ contain $2k$ channels, each point represents positive and negative excitation, the classification loss is obtained by calculating softmax. $A_{w \times h \times 4k}^{reg}$ contain $2k$ channels, each point represents d_x, d_y, d_w, d_h referring to the difference between the estimated value and the ground-truth.

The loss function is defined as

$$L_{reg} = \sum_{i=0}^3 \text{smooth}_{L_1}(\delta[i], \sigma) \quad (3)$$

$$\begin{aligned} \delta[0] &= (T_x - A_x)/A_w \\ \delta[1] &= (T_y - A_y)/A_h \\ \delta[2] &= \ln(T_w/A_w) \\ \delta[3] &= \ln(T_h/A_h) \end{aligned} \quad (4)$$

where A_x, A_y, A_w and A_h are the coordinates of the center point and length and width of the anchor point frame. T_x, T_y, T_w and T_h are the center coordinates and dimensions of the real box. The loss smooth_{L_1} in formula 3 is defined as

$$\text{smooth}_{L_1} = \begin{cases} 0.5\sigma^2 x^2 & \text{if } |x| < 1/\sigma^2 \\ |x| - 1/2\sigma^2 & \text{if } |x| \geq 1/\sigma^2 \end{cases} \quad (5)$$

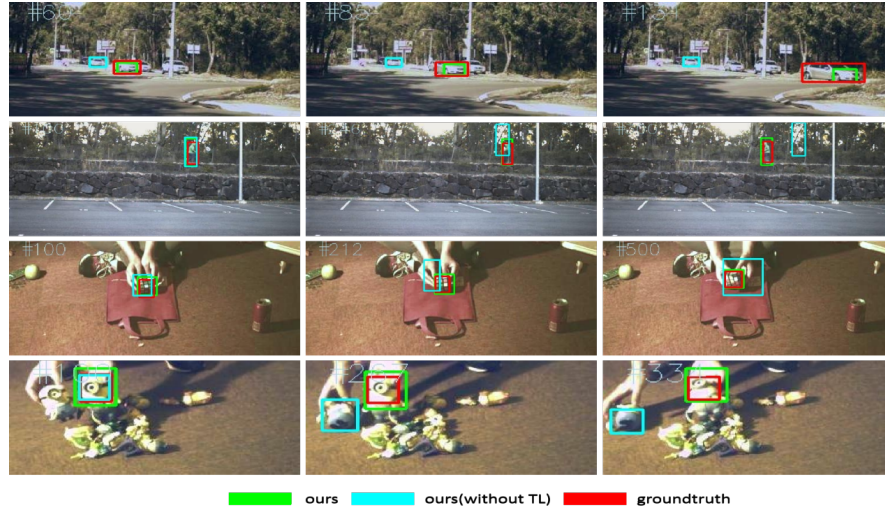


Fig. 1: Several tracking results.

The final loss function is

$$\text{loss} = L_{cls} + \lambda L_{reg} \quad (6)$$

where λ is the hyperparameter of balancing classification loss and regression loss, and L_{cls} is the cross entropy loss.

B. Hyperspectral band selection

In order to reduce the complexity of the tracking process and the redundancy of the HSVs, whose intrinsic feature needs to be obtained via feature dimension reduction methods, including feature selection [38] and feature extraction [11]. The former refers to selecting features with good classification performance, i.e., band selection of HSVs. Feature extraction aims to map features from the original space to a new low-dimensional space, such as PCA, linear discriminant analysis and singular value decomposition. Yet, it needs to give new definition of feature attributes. Therefore, this paper utilizes genetic algorithm [39] to retain most of the original information.

In general, the correlation between retained bands should not be too high. The diagram of correlation coefficient and the operation of band selection are shown in Fig. 2.

As can be seen from Fig. 2, the matrix diagram of the correlation coefficient indicates that the initial images can be divided into several groups. Moreover, the correlation coefficient of intra-group images is high, and the inter-group correlation is low inversely. Therefore, the redundancy of original information should be reduced.

In view of the large amount of multispectral video data, the calculation of information entropy between bands is also time-consuming. Based on the above analysis, genetic optimization algorithm [40], [41] is introduced to quickly obtain three bands with a maximum entropy value. Given the initial image $D =$

$[d_1; d_2; \dots; d_{16}] \in R^{m \times n \times 16}$, the joint information entropy H of band d_i d_j and d_k can be expressed as

$$H(d_i, d_j, d_k) = - \sum_{I,J,K=0}^{255} P_{I,J,K} \log_2 P_{I,J,K} \quad (7)$$

where, $P_{I,J,K}$ is the joint probability between pixel value I of band d_i , pixel value J of band d_j and pixel value K of band d_k .

According to the above analysis, the joint entropy can be seen as the fitness function of the discrete genetic algorithm. Besides, the band number group is used as the chromosome in the genetic algorithm. In order to determine suitable bands, selection, crossover and mutation are iteratively implemented, which is show in Fig. 2. The reason why we obtain three bands is according to the correlation analysis. Besides, the basic deep network needs an input with three spectrums.

To sum up, bands are selected based on the joint entropy, which takes into account the information reduction and the execution efficiency. That is to say, increasing the number of bands for the given spectral data will result in information redundancy.

C. Transfer learning

Transfer learning (TL) refers to the transfer of a model trained in one task to another. In some application scenarios, there will be some contradictions in the training of deep learning model, such as the contradiction between big data and less annotation, and the data annotation cost is high.

For Single object tracking, DL models are usually trained on large data sets without transfer learning. However, for small data sets, pre-trained operation on large data is needed, and then the trained model is utilized to fine-tune on the specific small target data. Existing hyperspectral trackers tend to use RGB data for pre-training, including ImageNet [37], COCO [42], GOT-10K [33], and so on, due to limited hyperspectral videos.

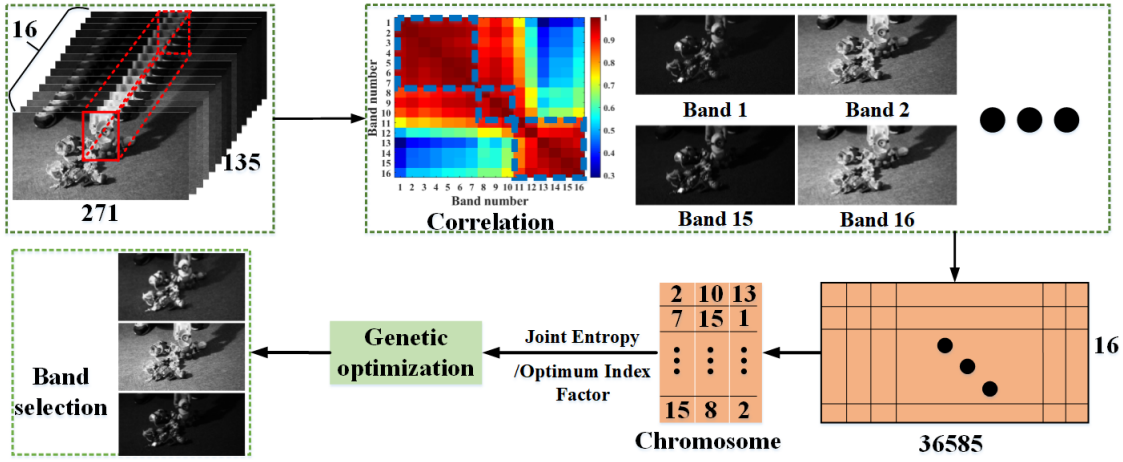


Fig. 2: Band selection for the initial toy1 image.

Using RGB data for pre-training will lose the semantic information of hyperspectral data. Besides, the hyperspectral data can not be combined with transfer learning directly, for the reason that the data are multi-channel. Therefore, the hyperspectral band selection module realized in this paper is necessary. It transforms the hyperspectral image into three bands, which is convenient for the TL model to obtain features with strong classification performance.

To sum up, the flow chart of the proposed algorithm is shown in Fig. 3

III. EXPERIMENTAL ANALYSIS

The experiment used the GOT-10K dataset and 35 hyperspectral videos provided by the competition website. The proposed BS-SiamRPN method was implemented in Anaconda3 and ran at 70 frame per second on a PC with 32G RAM and Nvidia GPU(3080). To better analyze the performance of the tracker, we compared it with several state-of-the-art algorithms, including the SiamRPN method [2] on RGB data, the BaeNet method [20], and the proposed BS-SiamRPN method without TL [43]. Qualitative results of existing trackers are shown in Figs. 4-7.

In this paper, we train the SiamRPN [33] network using GOT-10K [24] training data set. Stochastic Gradient Descent (SGD) method was used to pretrain the network. In a total of 50 epochs, the learning rate warmed up from 5×10^{-4} to 1×10^{-3} in the first 5 epochs and then decayed to 5×10^{-4} in the next 45 epochs. Besides, the training data set provided by the competition website is utilized to conduct transfer learning for the pre-training model. In 20 epochs, the learning rate attenuated from 1×10^{-4} to 5×10^{-5} . The SGD method is used to transfer the network.

A. Qualitative comparison

Four groups of video data, namely, *Basketball*, *Car3*, *Coke* and *Paper*, were processed to verify the tracking performance of the proposed algorithm. Fig. 4- 7 show partial

tracking results of the proposed algorithm and BaeNet algorithm on hyperspectral videos, as well as tracking results of SiamRPN on RGB videos.

In Fig. 4, the target in video *basketball* is small and moves fast. At the initial stage, both BaeNet and the algorithm in this paper can detect the target. However, in the subsequent rapid movement of the target, BaeNet loses the target and cannot re-detect the target for a long period of time until it returns to the place where the target is lost. By contrast, the algorithm in this paper can detect small and fast-moving objects.

In Fig. 5, the target in *car3* is similar to many objects in the background, resulting in the wrong object detected by BaeNet, and the target cannot be detected in the whole process. The algorithm in this paper does not lose the target. Besides, in Fig. 6, the target in *coke* is similar in color to the local background, and rotates in a non-plane way over a period of time. In the initial stage, both the BaeNet method and the proposed method can detect the target. Yet, in the subsequent detection, the former detection frame cannot closely fit the rotating target.

In Fig. 7, the target paper has a similar color to the white background. In the initial stage, both BaeNet and the algorithm in this paper can detect the target. In the subsequent target movement, BaeNet can only detect a part of the object, and the algorithm in this paper can completely detect the target. SiamRPN which relies on RGB video to extract features completely loses its target.

B. Quantitative comparison

In this section, we compare four methods quantitatively using precision plot and success plot. Precision plot shows the percentages of successful frames whose distance between the center of predicted bounding box and the center of ground-truth is smaller than a threshold (in pixels), which reflects the accuracy of the center positioning. Moreover, success plot shows the percentages of successful frames whose overlap ratio is larger than a threshold varied from 0 to 1, which reflects the accuracy of scale selection. The Area Under Curve

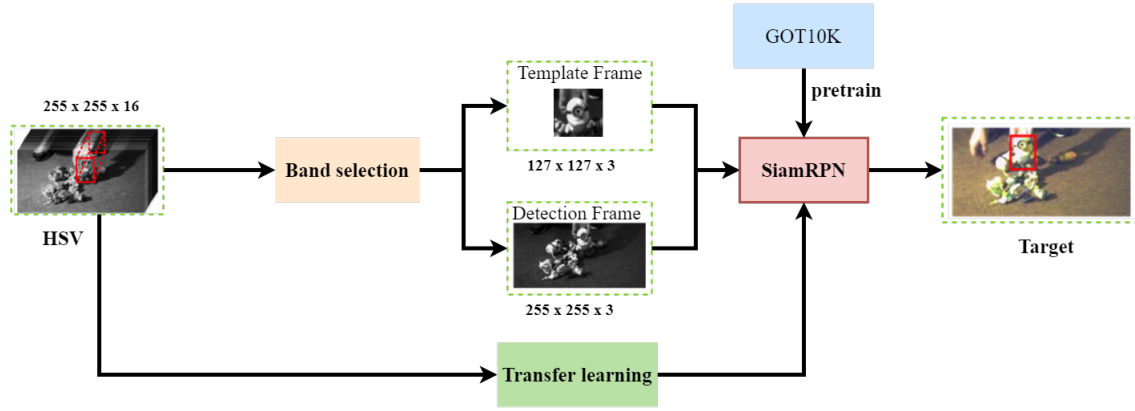


Fig. 3: The flow chart of the proposed algorithm.

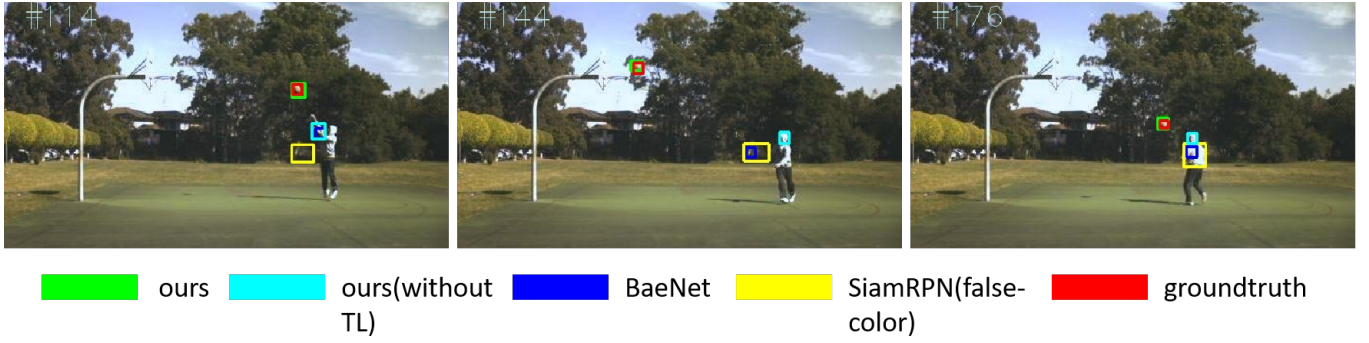


Fig. 4: Qualitative results on the *basketball* sequence.

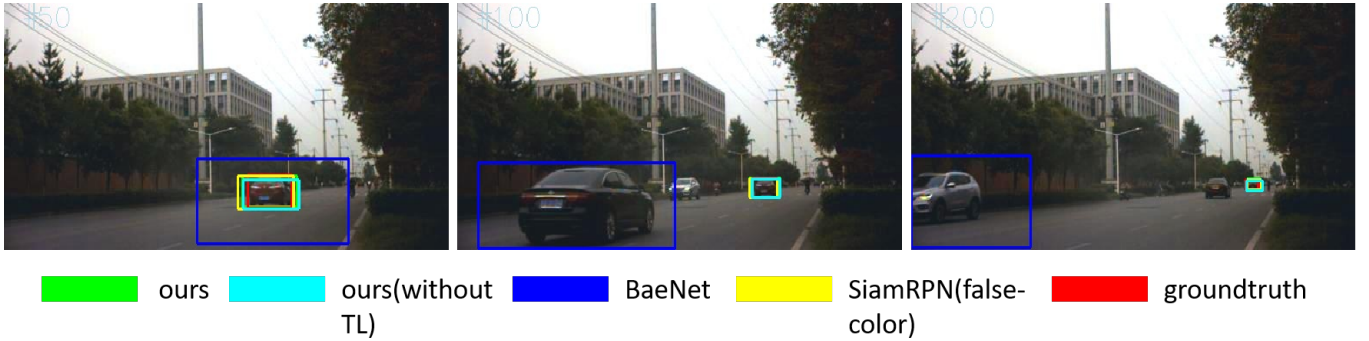


Fig. 5: Qualitative results on the *car3* sequence.

TABLE I: AUC comparison

Algorithm	Video type	AUC
SiamRPN	false-color	0.483
BaeNet	HSV	0.606
Ours(Without TL)	HSV	0.468
Ours	HSV	0.526

(AUC) of each plot is an overall evaluation measure to analyze the performance of all trackers. Table. I shows the AUC comparison of the strong benchmark deep learning based RGB trackers and hyperspectral trackers.

Fig. 8 shows the quantitative results for all sequences. It can be seen that the proposed method gets the great AUC score in both plots (0.8332 in precision plot and 0.526 in success plot). The results show that due to the characteristics of occlusion, similar background, fast motion and low resolution of the videos in the dataset, SiamRPN cannot accurately locate the target only by relying on the features obtained from RGB videos. In the algorithm without transfer learning proposed in this paper, the input hyperspectral data cannot be fully understood, so it is easy to lead to tracking failure. After transfer learning, our algorithm will learn the semantic features of multi-spectrum, so it achieves good results. BaeNet used the band selection network to send the band group into the

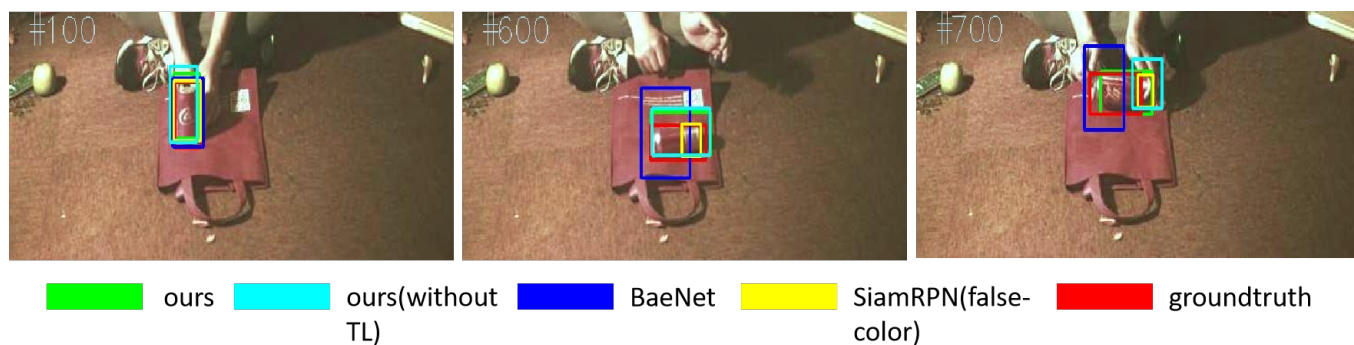


Fig. 6: Qualitative results on the *coke* sequence.

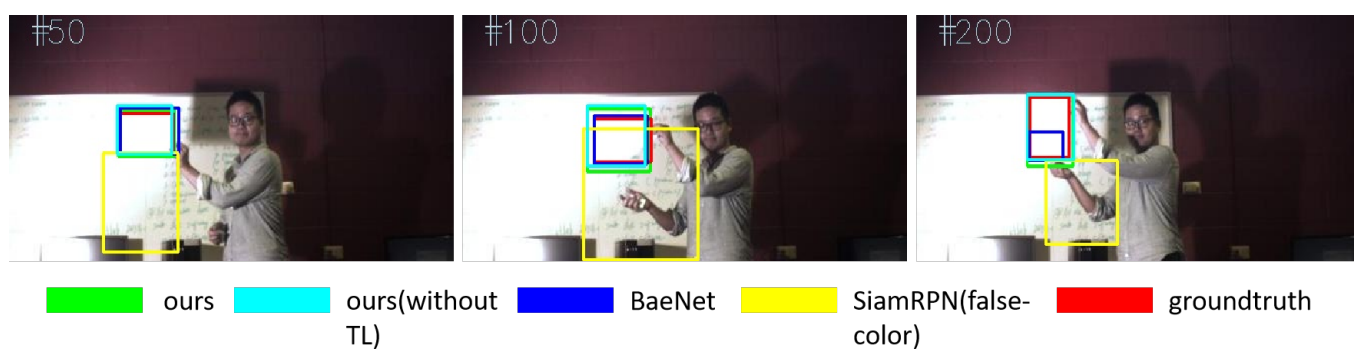
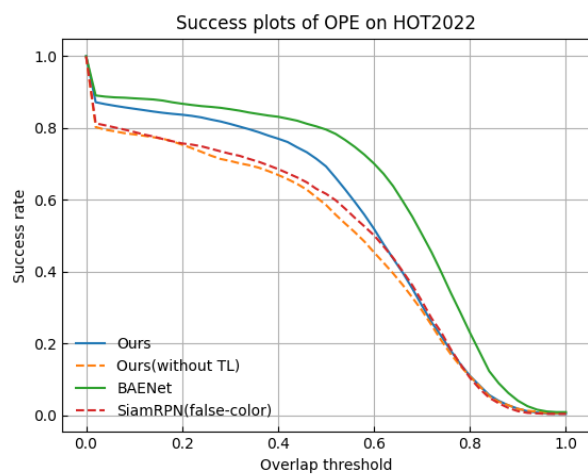
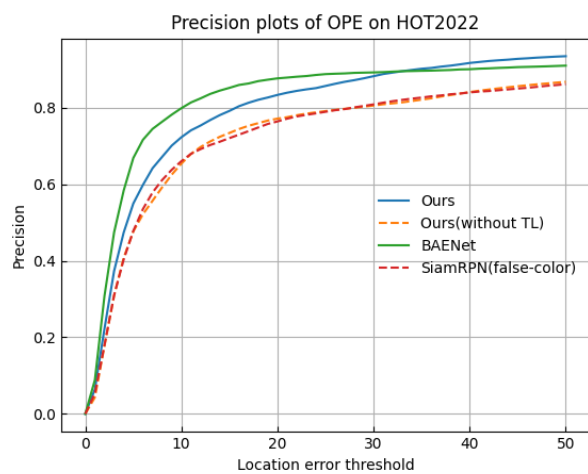


Fig. 7: Qualitative results on the *paper* sequence.



(a) Success plots



(b) Precision plots

Fig. 8: Precision plot of existing algorithms.

VITAL tracker for tracking. Although better performance was achieved, our algorithm was better in the balance of speed and accuracy.

IV. CONCLUSIONS

In this paper, an algorithm based on band selection and siamese networks, is proposed for hyperspectral video target tracking. Hyperspectral information improves the performance of tracking algorithms to distinguish objects with similar color features from background information. In addition, a better model is trained on a recent visible GOT database. Besides, transfer learning helps to extract semantic information of hyperspectral video, thus improving the performance of tracker. Experimental results verify the superior performance of the proposed algorithm, and the area under the curve of the success rate graph reaches 0.52, which lays a solid result for the subsequent target detection and recognition.

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REFERENCES

- [1] K. Qian, J. Zhou, F. Xiong, H. Zhou, and J. Du, "Object tracking in hyperspectral videos with convolutional features and kernelized correlation filter," in *International Conference on Smart Multimedia*, pp. 308–319, Springer, 2018.
- [2] N. Wang and D.-Y. Yeung, "Learning a deep compact image representation for visual tracking," *Advances in neural information processing systems*, vol. 26, pp. 71–81, 2019.
- [3] L. Wang, W. Ouyang, X. Wang, and H. Lu, "Visual tracking with fully convolutional networks," in *Proceedings of the IEEE international conference on computer vision*, pp. 3119–3127, 2015.
- [4] L. Bertinetto, J. Valmadre, J. F. Henriques, A. Vedaldi, and P. H. Torr, "Fully-convolutional siamese networks for object tracking," in *European conference on computer vision*, pp. 850–865, Springer, 2016.
- [5] H. Nam and B. Han, "Learning multi-domain convolutional neural networks for visual tracking," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4293–4302, 2016.
- [6] X. Li, Q. Liu, N. Fan, Z. He, and H. Wang, "Hierarchical spatial-aware siamese network for thermal infrared object tracking," *Knowledge-Based Systems*, vol. 166, pp. 71–81, 2019.
- [7] P. Li, B. Chen, W. Ouyang, D. Wang, X. Yang, and H. Lu, "Gradnet: Gradient-guided network for visual object tracking," in *Proceedings of the IEEE/CVF International conference on computer vision*, pp. 6162–6171, 2019.
- [8] C. Xiu and Z. Chai, "Target tracking based on the cognitive associative network," *IET Image Processing*, vol. 13, no. 3, pp. 498–505, 2019.
- [9] D. S. Bolme, J. R. Beveridge, B. A. Draper, and Y. M. Lui, "Visual object tracking using adaptive correlation filters," in *2010 IEEE computer society conference on computer vision and pattern recognition*, pp. 2544–2550, IEEE, 2010.
- [10] J. F. Henriques, R. Caseiro, P. Martins, and J. Batista, "High-speed tracking with kernelized correlation filters," *IEEE transactions on pattern analysis and machine intelligence*, vol. 37, no. 3, pp. 583–596, 2014.
- [11] K. Zhang, L. Zhang, Q. Liu, D. Zhang, and M.-H. Yang, "Fast visual tracking via dense spatio-temporal context learning," in *European conference on computer vision*, pp. 127–141, Springer, 2014.
- [12] M. Danelljan, G. Hager, F. Shahbaz Khan, and M. Felsberg, "Learning spatially regularized correlation filters for visual tracking," in *Proceedings of the IEEE international conference on computer vision*, pp. 4310–4318, 2015.
- [13] L. Zhang and P. N. Suganthan, "Robust visual tracking via co-trained kernelized correlation filters," *Pattern Recognition*, vol. 69, pp. 82–93, 2017.
- [14] H. Kiani Galoogahi, A. Fagg, and S. Lucey, "Learning background-aware correlation filters for visual tracking," in *Proceedings of the IEEE international conference on computer vision*, pp. 1135–1143, 2017.
- [15] H. Liu and B. Li, "Target tracker with masked discriminative correlation filter," *IET Image Processing*, vol. 14, no. 10, pp. 2227–2234, 2020.
- [16] X. Lan, Z. Yang, W. Zhang, and P. C. Yuen, "Spatial-temporal regularized multi-modality correlation filters for tracking with re-detection," *ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM)*, vol. 17, no. 2, pp. 1–16, 2021.
- [17] X. Lu, J. Li, Z. He, W. Wang, and H. Wang, "Distracter-aware tracking via correlation filter," *Neurocomputing*, vol. 348, pp. 134–144, 2019.
- [18] S. Moorthy, J. Y. Choi, and Y. H. Joo, "Gaussian-response correlation filter for robust visual object tracking," *Neurocomputing*, vol. 411, pp. 78–90, 2020.
- [19] M. Danelljan, F. Shahbaz Khan, M. Felsberg, and J. Van de Weijer, "Adaptive color attributes for real-time visual tracking," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 1090–1097, 2014.
- [20] N. Wang, W. Zhou, Q. Tian, R. Hong, M. Wang, and H. Li, "Multi-cue correlation filters for robust visual tracking," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4844–4853, 2018.
- [21] J. Kong, Y. Ding, M. Jiang, and S. Li, "Collaborative model tracking with robust occlusion handling," *IET Image Processing*, vol. 14, no. 9, pp. 1701–1709, 2020.
- [22] P. Voigtlaender, J. Luiten, P. H. Torr, and B. Leibe, "Siam r-cnn: Visual tracking by re-detection," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6578–6588, 2020.
- [23] B. Wei, H. Chen, Q. Ding, and H. Luo, "Siamoon: Siamese object-aware network for real-time target tracking," *Neurocomputing*, vol. 471, pp. 161–174, 2022.
- [24] B. Li, J. Yan, W. Wu, Z. Zhu, and X. Hu, "High performance visual tracking with siamese region proposal network," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 8971–8980, 2018.
- [25] K. Zhang, Q. Liu, W. Yi, S. Member, and IEEE, "Robust visual tracking via convolutional networks," *IEEE Transactions on Image Processing*, vol. 25, no. 4, pp. 1779–1792, 2016.
- [26] B. Li, W. Wu, Q. Wang, F. Zhang, J. Xing, and J. Yan, "Siamrpn++: Evolution of siamese visual tracking with very deep networks," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 4282–4291, 2019.
- [27] D. Guo, J. Wang, Y. Cui, Z. Wang, and S. Chen, "Siamcar: Siamese fully convolutional classification and regression for visual tracking," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 6269–6277, 2020.
- [28] Z. Zhang, H. Peng, J. Fu, B. Li, and W. Hu, "Ocean: Object-aware anchor-free tracking," in *European Conference on Computer Vision*, pp. 771–787, Springer, 2020.
- [29] Z. Li, F. Xiong, J. Zhou, J. Wang, J. Lu, and Y. Qian, "Bae-net: A band attention aware ensemble network for hyperspectral object tracking," in *2020 IEEE International Conference on Image Processing (ICIP)*, pp. 2106–2110, IEEE, 2020.
- [30] H. Van Nguyen, A. Banerjee, and R. Chellappa, "Tracking via object reflectance using a hyperspectral video camera," in *2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition-Workshops*, pp. 44–51, IEEE, 2010.
- [31] F. Xiong, J. Zhou, and Y. Qian, "Material based object tracking in hyperspectral videos," *IEEE Transactions on Image Processing*, vol. 29, pp. 3719–3733, 2020.
- [32] Y. Song, C. Ma, X. Wu, L. Gong, L. Bao, W. Zuo, C. Shen, R. W. Lau, and M.-H. Yang, "Vital: Visual tracking via adversarial learning," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 8990–8999, 2018.

- [33] L. Huang, X. Zhao, and K. Huang, "Got-10k: A large high-diversity benchmark for generic object tracking in the wild," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 43, no. 5, pp. 1562–1577, 2019.
- [34] W. Luo, X. Li, W. Li, and W. Hu, "Robust visual tracking via transfer learning," in *2011 18th IEEE International Conference on Image Processing*, pp. 485–488, IEEE, 2011.
- [35] J. Valmadre, L. Bertinetto, J. Henriques, A. Vedaldi, and P. H. Torr, "End-to-end representation learning for correlation filter based tracking," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2805–2813, 2017.
- [36] Q. Guo, W. Feng, C. Zhou, R. Huang, L. Wan, and S. Wang, "Learning dynamic siamese network for visual object tracking," in *Proceedings of the IEEE international conference on computer vision*, pp. 1763–1771, 2017.
- [37] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, *et al.*, "Imagenet large scale visual recognition challenge," *International journal of computer vision*, vol. 115, no. 3, pp. 211–252, 2015.
- [38] L. Xijun and L. Jun, "An adaptive band selection algorithm for dimension reduction of hyperspectral images," in *2009 International Conference on Image Analysis and Signal Processing*, pp. 114–118, IEEE, 2009.
- [39] H. Zhao, L. Bruzzone, R. Guan, F. Zhou, and C. Yang, "Spectral-spatial genetic algorithm-based unsupervised band selection for hyperspectral image classification," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 59, no. 11, pp. 9616–9632, 2021.
- [40] D. Akbari, "Improved neural network classification of hyperspectral imagery using weighted genetic algorithm and hierarchical segmentation," *IET Image Processing*, vol. 13, no. 12, pp. 2169–2175, 2019.
- [41] P. J. Angeline, G. M. Saunders, and J. B. Pollack, "An evolutionary algorithm that constructs recurrent neural networks," *IEEE transactions on Neural Networks*, vol. 5, no. 1, pp. 54–65, 1994.
- [42] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *European conference on computer vision*, pp. 740–755, Springer, 2014.
- [43] J. Gao, H. Ling, W. Hu, and J. Xing, "Transfer learning based visual tracking with gaussian processes regression," in *European conference on computer vision*, pp. 188–203, Springer, 2014.